

Solar
catalogue
2025

POLYCAB

EVERYTHING UNDER THE SUN



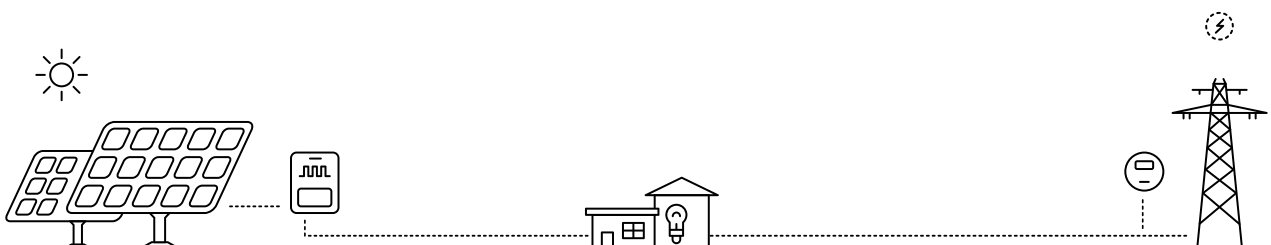
Grid Tie Inverter | PV Module | Solar DC Cables

POLYCAB



CONTENT

Polycab Introduction	01
Single Phase Inverter	03
Three Phase Inverter	05
PV Module	13
DC Cable	17



INTRODUCTION

Polycab is the country's largest manufacturer of wires and cables, manufacturing around 3.9 million kilometers of cables every year. Underpinning our leadership position are our solid business fundamentals, which include multi-location manufacturing with a high degree of backward integration, a comprehensive product portfolio, strong brand positioning, robust distribution network, and experienced management. Polycab's widest range of wires & cables helps the company bond with millions of satisfied customers, riding on key differentiators like product innovation, superior quality, and ready availability. Our clientele includes market leaders in sectors like utilities, power generation, transmission and distribution, petroleum and oil refineries, original equipment manufacturers, EPC contractors, steel, metal, cement, chemicals, atomic energy, railways, and nuclear power industries amongst others.

Apart from a stellar lineup of wires and cables, we have made inroads into the highly competitive FMEG market, with products like Fans, LED Lighting and Luminaires, Switches and Switchgears, Home Appliances, Solar Products and Conduits & Accessories. Polycab's corporate advantage includes its extensive base of expertise, proven technological capabilities, and comprehensive skills of its human resources.

Solar-the Infinite Source Of Power

The sun provides us with ample energy than we could ever use, and no one can monopolies the sunlight. Sun light is free and can be used to convert into electrical energy which is referred as Solar PV system. Solar electricity is green renewable energy and doesn't release any harmful carbon dioxide or other pollutants. A typical 3 kW home solar PV system could save around 3 tons of carbon per year.

With the continuously increasing demand for electric power, the significantly high price of oil and the growing concern for the environment, many businesses are in the process of implementing alternative sources of energy. Among the renewable energy sources, solar energy is a sustainable choice and that can be used in various applications. Many businesses are now extracting this alternative source of energy, hoping to benefit from its numerous advantages.

To make an ecological awareness and safe use of renewable energy Polycab has brought complete Solar energy solution in Indian and overseas market. Polycab Solar equipment meet the high expectation that are demanded from the Solar system.

Polycab has brought the environmentally friendly E-Beam Technology that meets the demand of sustainable product in line with worldwide market trends and ecological awareness.

Polycab has a comprehensive product range in Solar PV system. The products are manufactured in latest state of the art machines and tested in well-equipped laboratories. These are highly suitable in rough climatic condition as well as guaranteed for more than 25 years of use.

We at Polycab ventured into Solar in 2012 with manufacturing of Solar DC Cables. International accreditation from TUV Rheinland was secured for our Solar DC cables subsequently, initially for 2Pfg 1169/08.2007 standard and then for EN 50618 for sizes 1.5sq. mm to 300sq.mm complying also to IEC 62930.

We have successfully supplied Solar DC as well as AC Cables to large EPC players, Distributors pan India as well as to many of our International Clients all over the Globe. Repeat orders have been forthcoming out of confidence on our product quality and supply capabilities.

Extending our foray into Solar field we added Solar On-Grid Inverters in our Product Basket in 2016. Polycab Solar Grid-Tie String Inverters have already captured the hearts of Solar Roof-Top System Integrators pan India through product performance and prompt after Sales-Services provided by Polycab. Polycab On-Grid Inverters are IEC / BIS Certified with all relevant applicable standards for the full range of Inverters. Polycab has established Solar Grid Tie Inverter manufacturing setup and have started offering MAKE IN INDIA inverters from **2 kW to 125 kW** capacity, catering to Residential, Commercial & Industrial solar projects from KW to MW scale. These inverters designed with highest reliability, efficiency to make compatible with latest module technology and are IEC & BIS certified.

Our success story of On-Grid Inverters in short span of **9 years** is worth mentioning. We have already supplied **1.5 GW** in capacity and **1Lac+ Inverters** in quantity. All these Inverters are already installed and running successfully in the field. We are sure to capture good market share.

We have also added **Solar PV Modules and Solar BOS Products like Solar DC MCBs, MC4 connectors, Cable Tray, Lugs and Glands** in our product basket to achieve our goal to become a one stop shop for all the major components needed in Solar Projects.

With our strong base, large network of branches, warehouses, and distributors across India, we have been extending our best sales and service support to our valued clients and end users.

With excellent performance and better generation Polycab Solar Products have been the most preferred choice amongst many EPC companies, System Integrators and End Users.

System Integrator is now nearing reality and we are now poised to offer all our products to International Markets.

NEW

POLYCAB

EXCLUSIVE FOR RESIDENTIAL SMALL ROOFTOP SOLAR PROJECTS

Single Phase

PSIS2K0SM1R2, PSIS3K0SM1R2,
PSIS3K6SM1R2, PSIS4K0SM1R2,
PSIS4K6SM1R2, PSIS5K0SM1R2,
PSIS5K5SM1R2, PSIS6K0SM1R2



LEADING FEATURES



Superior Efficiency

Maximum efficiency 97.3%
String Current 20A.
Compatible with High Power TOPCon Modules.

High Reliability

IP65 waterproof and dustproof, C5 anti corrosion
DC / AC Inbuilt surge protections.
Compatible with wide power grid voltage and high harmonic power grid environment

Intelligent Maintenance

App quick commissioning
Remote configuration and upgrade

Single Phase

2 kW | 3 kW | 3.6 kW | 4 kW | 4.6 kW | 5 kW | 5.5 kW | 6 kW

TECHNICAL SPECIFICATIONS

MODEL	PSIS2K0 SMIR2	PSIS3K0 SMIR2	PSIS3K6 SMIR2	PSIS4K0 SMIR2	PSIS4K6 SMIR2	PSIS5K0 SMIR2	PSIS5K5 SMIR2	PSIS6K0 SMIR2
RATING	2 kW	3 kW	3.6 kW	4 kW	4.6 kW	5 kW	5.5 kW	6 kW
INPUT (PV)								
Max. Usable Power	150%							
Max. DC Voltage	500			600				
MPPT Range	50-490			70-550				
MPPT Range(Full load)	110-430	160-430	190-430	210-500	240-500	260-500	290-500	320-500
Nominal Input Voltage	360							
Max. DC Current	20			20				
Max. Short Circuit Current	30			30				
MPP Tracker No.	1			1				
String No.	1			1				
OUTPUT (GRID)								
Nom. Power (Output)	2,000 W	3,000 W	3,600 W	4,000 W	4,600 W	5,000 W	5,500 W	6,000 W
Maximum Power (Output)	2,200 VA	3,300 VA	3,600 VA	4,400 VA	5,060 VA	5,500 VA	6,000 VA	6,000 VA
Apparent Power (Output)	2,200 W	3,300 W	3,600 W	4,400 W	5,060 W	5,500 W	6,000 W	6,000 W
Nominal Voltage	220 V / 230 V / 240 V							
Rated grid frequency	50 Hz / 60 Hz							
Maximun Current (Output)	10.0 A	15.0 A	16.0 A	20.0 A	23.0 A	25.0 A	27.3 A	27.3 A
Power Factor Adjustment Range	0.8 leading to 0.8 lagging							
THD	<3% (Rated Power)							
EFFICIENCY								
Max. Efficiency(PV to Grid)	97.3%			97.2%				
Eur. Efficiency(PV to Grid)	95.9%	95.9%	96.3%	96.0%		96.2%	96.4%	96.4%
GENERAL								
Temp. Range	-25°~60°							
Max. operation altitude	4000m							
Topology	Transformerless							
Protection	IP65							
Noise emission	30 dB							
Humidity	0-100%							
Cooling	Natural convection							
HMI & COM								
Display	LED+LCD, APP							
Communication interface	RS485/RS232, WiFi/LAN (Optional)							
PROTECTION								
Anti-islanding protection	YES							
AC over-current protection	YES							
AC short-circuit protection	YES							
AC over-voltage protection	YES							
PV Reverse Polarity Protection	YES							
Residual Current Monitoring	YES							
Surge Arrestor	DC/AC							
Insulation detection	YES							
MECHANICAL								
W x H x D	280mm*336mm*130mm			320mm*344mm*137mm				
Weight	5 KG			9 KG				
DC switch	YES							
CERTIFICATION								
Anti-Islanding Protection	IS 16169:2014							
Safety	IS 16221 (PART 2):2015							
WARRANTY	7 Years							

Note : ①The range of output voltage and frequency may vary depending upon different grid codes.
 ②Specifications are subject to change without advance notice.

NEW

POLYCAB

SUITABLE FOR RESIDENTIAL & COMMERCIAL ROOFTOP SOLAR PROJECTS

Three phase

PSIT5KSM1R2

PSIT6KSM1R2

PSIT8KSM1R2

PSIT10KSM1R2



Leading Features



Superior Efficiency

Maximum efficiency 97.8%
Maximum String Current , compatible with 600W+ modules
150% PV configuration, 110% output overload

High Reliability

IP66 waterproof and dustproof, C5 anti corrosion
DC / AC surge protections.
Compatible with wide power grid voltage and high harmonic power grid environment

Intelligent Maintenance

App quick commissioning
Remote configuration and upgrade

Three Phase

PSIT-5K-SMIR2 | PSIT-6K-SMIR2 | PSIT-8K-SMIR2 | PSIT-10K-SMIR2

TECHNICAL SPECIFICATIONS				
MODEL	PSIT-5K SMIR2	PSIT-6K SMIR2	PSIT-8K SMIR2	PSIT-10K SMIR2
Rating	5 kW	6 kW	8 kW	10 kW
INPUT (PV)				
Max. Input Voltage	1100V			
Max. PV configuration	150%			
Rated Input Voltage	620V			
Max. Input Current	20A			
Max. Short Circuit Current	30A			
Start Input Voltage	180V			
MPPT Operating Range	160V -1000V			
Max. Number of PV Strings	1			
No. of MPPTs	1			
OUTPUT (GRID)				
Rated AC Active Power	5,000W	6,000W	8,000W	10,000W
Max. AC Apparent Power	5,500VA	6,600VA	8,800VA	11,200VA
Max. AC Active Power (PF=1)	5,500W	6,600W	8,800W	11,200W
Max. AC Output Current	3*8.4A	3*10.1A	3*13.4A	3*17A
Rated AC Voltage	380V/400V/415V, 3W+N+PE			
AC Voltage Range¹	260V- 510V (Adjustable)			
Rated Grid Frequency	50Hz/60Hz			
Grid Frequency Range²	45Hz-55Hz/55Hz -65Hz (Adjustable)			
THDI	<3%@Rated Power			
DC Current Injection	<0.5%@Rated Current			
Power Factor	>0.99 Rated power (Adjustable 0.8 Leading – 0.8 Lagging)			
EFFICIENCY				
Max. Efficiency	97.8%	97.8%	97.8%	97.8%
European Efficiency	96.2%	96.5%	96.9%	97.1%
PROTECTION				
DC switch	Support			
Anti-islanding protection	Support			
AC overcurrent protection	Support			
AC short circuit protection	Support			
DC reverse connection	Support			
Surge Arrester	AC Type III / DC Type III			
Insulation detection	Support			
Leakage current protection	Support			
GENERAL				
Topology	Transformer-less			
IP Rating	IP66			
Night Self Consumption	<1W			
Cooling	Natural cooling			
Operating Tmp. Range	-25°C -60°C			
Relative Humidity Range	0 – 100%			
Max. Operating Altitude	4000m			
Noise	<30dB			
Dimensions (W*H*D)	398mm*460mm*190mm			
Weight	16.8 Kgs			
HMI & COM				
Display	Wireless & APP+LED, LCD			
Communication	Optional: WiFi/GPRS/RS485/LAN			
CERTIFICATION				
Anti-Islanding Protection	IS 16169:2014			
Safety	IS 16221 (PART 2):2015			
WARRANTY	7 Years			

Note :

①②The range of output voltage and frequency may vary depending upon different grid codes.
Specifications are subject to change without advance notice.

SUITABLE FOR RESIDENTIAL & COMMERCIAL ROOFTOP SOLAR PROJECTS

Three phase

PSIT-12K-SM3 , PSIT-15K-SM3 ,
PSIT-20K-SM4 , PSIT-25K-SM4
PSIT-30K-SM4



LEADING FEATURES



Superior Efficiency

Maximum efficiency 98.4%
Maximum String Current , compatible with 600W+ modules
150% PV configuration, 110% output overload

High Reliability

IP66 waterproof and dustproof, C5 anti corrosion
DC / AC surge protections.
Compatible with wide power grid voltage and high harmonic power grid environment

Intelligent Maintenance

App quick commissioning
Remote configuration and upgrade

Three Phase

PSIT-12K-SM3 | PSIT-15K-SM3 | PSIT-20K-SM4
PSIT-25K-SM4 | PSIT-30K-SM4

TECHNICAL SPECIFICATIONS					
MODEL	PSIT-12K SM3	PSIT-15K SM3	PSIT-20K SM4	PSIT-25K SM4	PSIT-30K SM4
Rating	12 kW	15 kW	20 kW	25 kW	30 kW
INPUT (PV)					
Max. Input Voltage	1100V				
Max. PV configuration	150%				
Rated Input Voltage	620V				
Max. Input Current	15A + 30A		30A + 30A		40A + 30A
Max. Short Circuit Current	20A + 40A		40A + 40A		50A + 37.5A
Start Input Voltage	180V				
MPPT Operating Range	160V -1000V				
Max. Number of PV Strings	3(1/2)		4(2/2)		
No. of MPPTs	2				
OUTPUT (GRID)					
Rated AC Active Power	12,000W	15,000W	20,000W	25,000W	30,000W
Max. AC Apparent Power	13,200VA	16,700VA	22,000VA	27,500VA	33,000VA
Max. AC Active Power (PF=1)	13,200W	16,700W	22,000W	27,500W	33,000W
Max. AC Output Current	3*20.2A	3*25.3A	3*33.7A	3*39.8A	3*50.2A
Rated AC Voltage	380V/400V/415V, 3W+N+PE				
AC Voltage Range¹	260V- 510V (Adjustable)				
Rated Grid Frequency	50Hz/60Hz				
Grid Frequency Range²	45Hz-55Hz/55Hz -65Hz (Adjustable)				
THDI	<3%@Rated Power				
DC Current Injection	<0.5%@Rated Current				
Power Factor	>0.99 Rated power (Adjustable 0.8 Leading - 0.8Lagging)				
EFFICIENCY					
Max. Efficiency	98.3%		98.4%		
European Efficiency	97.8%		98.0%		
PROTECTION					
DC switch	Support				
Anti-islanding protection	Support				
AC overcurrent protection	Support				
AC short circuit protection	Support				
DC reverse connection	Support				
Surge Arrester	AC Type III / DC Type III				
Insulation detection	Support				
Leakage current protection	Support				
GENERAL					
Topology	Transformerless				
IP Rating	IP66				
Night Self Consumption	<1W				
Cooling	Natural cooling				
Operating Tmp. Range	-25°C -60°C				
Relative Humidity Range	0 - 100%				
Max. Operating Altitude	4000m				
Noise	<30dB				
Dimensions (W*H*D)	398mm*460mm*190mm				
Weight	18.7 Kgs		20 Kgs		21.5 Kgs
HMI & COM					
Display	Wireless & APP+LED, LCD				
Communication	WiFi, RS485(Optional), GPRS(Optional)				
CERTIFICATION					
Grid Connection	IEC 61727				
Anti-Islanding Protection	IEC 62116				
Environmental Testing	IEC 60068-2 (1-2-14-30)				
Safety	IEC 62109-1, IEC 62109-2				
EMC	IEC 61000				
Efficiency Measurement	IEC 61683				
WARRANTY	7 Years				

Note :

①②The range of output voltage and frequency may vary depending upon different grid codes.
Specifications are subject to change without advance notice.

SUITABLE FOR COMMERCIAL & INDUSTRIAL ROOFTOP SOLAR PROJECTS

Three Phase

PSIT-40K-SM6

PSIT-50K-SM8

PSIT-60K-SM8



LEADING FEATURES



- Maximum Efficiency 98.2%
- Wide Voltage Range
- Supports 50% DC oversize,
- 3/4 MPPT design with precise MPPT algorithm
- THDi < 3%,
- IP66 Enclosure
- Intelligent Fan Cooling
- Intelligent string monitoring
- Smart I-V Curve Diagnosis
- Fuse free design to avoid hazard
- Type II SPD for both DC and A C side
- Integrated DC disconnect switches
- RS-485,Wi-Fi / GPRS monitoring interface
- Support "Y" type connection in DC side
- Supports aluminium wire access to reduce cost
- Free remote monitoring on Web Portal and Mobile App

Three Phase

PSIT-40K-SM6 | PSIT-50K-SM8 | PSIT-60K-SM8

TECHNICAL SPECIFICATIONS				
MODEL	PSIT-40K-SM6		PSIT-50K-SM8	PSIT-60K-SM8
RATING	40kW		50kW	60kW
INPUT (PV)				
Max. Input Voltage			1100V	
Max. PV configuration (STC)			150%	
Rated Input Voltage			620V	
Max. Input Current	40A+2*32A	40A+3*32A		40A+3*32A
Max. Short Circuit Current	50A+2*45A	50A+3*45A		50A+3*45A
Start Input Voltage			200V	
MPPT Operating Voltage Range			180-1000V	
Max. Number of PV Strings	2/2/2	2/2/2/2		2/2/2/2
No. of MPPTs	3	4		4
OUTPUT (GRID)				
Rated AC Active Power	40000W	50000W		60000W
Max. AC Apparent Power	44500VA	55600VA		66700VA
Max. AC Active Power (PF=1)	44500W	55600W		66700W
Max. AC Output Current	67.5A	84.3A		92A
Rated AC Voltage	380/400/415V, 3L/N/PE,3L/PE			
AC Voltage Range①	322V-520V			
Rated Grid Frequency	50Hz/60Hz			
Grid Frequency Range②	45~55Hz/55~65Hz			
THDI	3% (Rated Power)			
DC Current Injection	<0.5In			
Power Factor	>0.99 Rated power (Adjustable 0.8 leading - 0.8 Lagging)			
EFFICIENCY				
Max. Efficiency	98.2%	98.2%		98.2%
European Efficiency	97.8%	97.8%		97.8%
PROTECTIONS				
DC switch	YES			
Anti-islanding protection	YES			
AC overcurrent protection	YES			
AC short circuit protection	YES			
DC reverse protection	YES			
Surge Arrester	DC Type II / AC Type II			
Insulation detection	YES			
Leakage current protection	YES			
GENERAL				
Topology	Transformer-less			
IP Rating	IP66			
Night Self Consumption	<1W(standard)			
Cooling	Fan cooling			
Operating Temperature Range	-25°C-60°C			
Relative Humidity Range	0-100%			
Max. Operating Altitude	4000m			
Noise(typical)	<45dB	<45dB		<55dB
Dimensions (W*H*D)	635mm*530mm*224mm		635mm*530mm*233mm	
Weight	41.5kg	42kg		42 kg
HMI & COM				
Display	Wireless & APP+LED, LCD			
Communication	WiFi, RS485(Optional), GPRS(Optional)			
CERTIFICATION				
Grid Connection	IEC 61727			
Anti-Islanding Protection	IEC 62116			
Environmental Testing	IEC 60068-2 (1-2-14-30)			
Safety	IEC 62109-1, IEC 62109-2			
EMC	IEC 61000			
Efficiency Measurement	IEC 61683			
Warranty	7 Years			

Note : ①②The range of output voltage and frequency may vary depending upon different grid codes.
Specifications are subject to change without advance notice.

SUITABLE FOR COMMERCIAL & INDUSTRIAL SOLAR PROJECTS

Three Phase
PSIT-100K-AM10
PSIT-110K-AM10



LEADING FEATURES



- Maximum Efficiency 98.7%
- Wide Voltage Range
- Supports 50% DC overload
- 10 MPPT design with precise MPPT algorithm
- THDi < 2%, Low Harmonic Distortion
- IP66 for outdoor Installation
- Intelligent Fan Cooling
- Intelligent string monitoring
- Smart I-V Curve Diagnosis
- Fuse free design to avoid hazard
- Type II SPD for both DC and A C side
- Integrated DC disconnect switches
- RS-485, Wi-Fi / GPRS monitoring interface
- Support "Y" type connection in DC side
- Support aluminium wire access to reduce cost
- Free remote monitoring on Web Portal and Mobile App

Three phase

PSIT -100K - AM10 | PSIT -110K - AM10

TECHNICAL SPECIFICATIONS		
MODEL	PSIT-100K-AM10	PSIT-110K-AM10
Rating	100 kW	110 kW
INPUT (PV)		
Max. DC Input Voltage	1100 V	
Max. DC Input Current	30A *10	30A *10
MPPT Voltage Range	200 – 1000 V	
Recommended MPPT Operating Range	600 V	
No. of MPPT	10	10
Max. no. of Strings per MPPT	2	
OUTPUT (GRID)		
Rated Output Power	100 kW	110 kW
Max. Output Power	110 KVA	121 KVA
Max. Output Current	158.8A	174. 6A
Rated grid Voltage	400V	
Grid Voltage range	310 ~ 480Vac	
Rated Grid Frequency	50Hz / 60Hz	
Grid Frequency Range ²	45-55Hz / 55- 65Hz	
THD	<2% (Under the rated power)	
Power Factor	>0.99 (rated power) / 0.8 leading ~ 0.8 lagging	
DC Current Injection	<0.5% (Under the rated power)	
SYSTEM DATA		
Max. Efficiency	98.7%	98.7%
Euro. Efficiency	98.1%	98.1%
Humidity range	0-100% non-condensing	
Cooling type	Intelligent forced air cooling	
Temperature range	-25 ~ +60°C	
Power consumption at night	<1W	
Max. working altitude	4000m	
Display	LED / LCD	
Communication interface	WiFi, RS485(Optional), GPRS(Optional)	
PROTECTIONS		
DC reverse -polarity protection	Support	
Short circuit protection	Support	
Output over current protection	Support	
Output over voltage protection	Support	
Insulation resistance monitoning	Support	
Residual current detection	Support	
Surge protection	Support	
Grid monitoring	Support	
Islanding protection	Support	
Temperature protection	Support	
Integrated DC switch	Support	
MECHANICAL DATA		
Dimensions (WxHxD)	1050mm x 620mm x 333mm	
Weight	89kg	
Protection class	IP66	
CERTIFICATES		
Grid Connection standard	IEC 61727	
Anti - Islanding Protection	IEC 62116	
Environmental Testing	IEC 60068 – 2	
Efficiency Measurement	IEC 61683	
Safety standard	IEC 62109 - 1/2	
Electromagnetic compatibility	IEC 61000 – 6 – 2/4	
WARRANTY	7 Years	

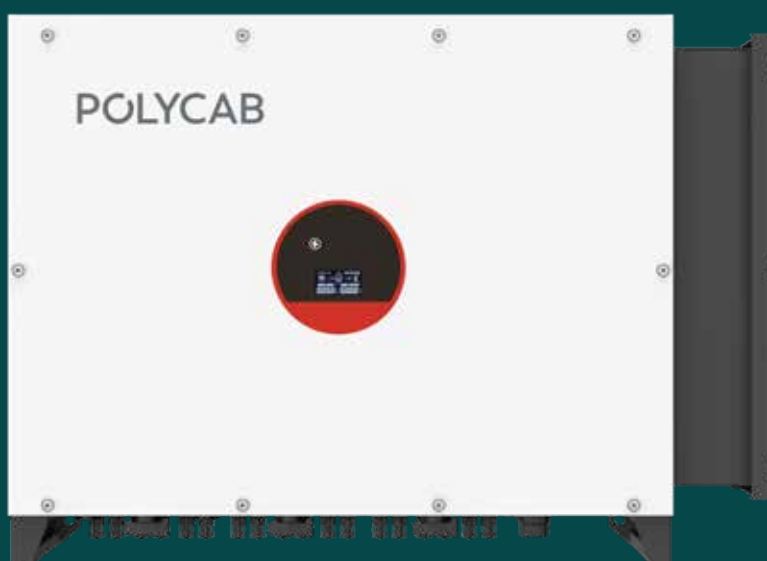
Note :
Specifications are subject to change without advance notice.

NEW

POLYCAB

SUITABLE FOR COMMERCIAL & INDUSTRIAL SOLAR PROJECTS

Three Phase
PSIT-125K-SM18



Leading Features



- Maximum Efficiency 98.2%
- Wide Voltage Range
- Supports 50% DC overload
- 9 MPPT design with precise MPPT algorithm
- THDi < 3%, Low Harmonic Distortion
- IP66 for outdoor Installation
- Intelligent Fan Cooling
- Intelligent string monitoring
- Smart I-V Curve Diagnosis
- Fuse free design to avoid hazard
- Type II SPD or MOV for both DC and AC side
- Integrated DC disconnect switches
- RS-485, Wi-Fi / GPRS monitoring interface
- Support "Y" type connection in DC side
- Support aluminium wire access to reduce cost
- Free remote monitoring on Web Portal and Mobile App

Three phase PSIT-125K-SM18

TECHNICAL SPECIFICATIONS	
MODEL	PSIT-125K-SM18
Rating	125 kW
INPUT (PV)	
Max. Input Power	150%
Max. DC Voltage	1100V
Startup Voltage	250 V
MPPT Range	200 - 1000 V
Rated Input Voltage	600 V
Max. DC Current	3*40A + 6*32A
Max. Short Circuit Current	3*50A + 6*45A
No. of MPPTs	9
No. of Strings	18 (9*2)
OUTPUT (GRID)	
Rated AC Power	125,000 W
Max. AC Active Power (PF=1)	125,000 W
Max. AC Apparent Power	125,000 VA
Nominal Voltage	400V, 3W+N+PE
AC Voltage Range*	322 - 520 V (adjustable)
Rated Grid Frequency	50 / 60 Hz
Grid Frequency Range **	45-55Hz / 55-65Hz (adjustable)
Maximum Current (Output)	3 * 181A
Power Factor	> 0.99 Rated Power (Adjustable 0.8 leading to 0.8 lagging)
THDi	< 3% (Rated Power)
EFFICIENCY	
Max. Efficiency	98.2%
Eur. Efficiency	98%
PROTECTIONS	
DC Switch	YES
Anti-islanding protection	YES
AC over-current protection	YES
AC short-circuit protection	YES
AC over-voltage protection	YES
DC Reverse Polarity Protection	YES
Surge Protection	DC Type II / AC Type II
Insulation detection	YES
Leakage Current Protection	YES
DC String Monitoring	YES
GENERAL	
Topology	Transformerless
IP Rating	IP66
Night Self Consumption	< 10W
Cooling	Fan Cooling
Operating Temp. Range	-25 ~ +60° C
Max. Operating Altitude	4000m
Humidity	0-100%
Noise	65dB
HMI & COM	
Display	LED+LCD, APP
Communication interface	WiFi, RS485/GPRS (Optional)
MECHANICAL DATA	
Dimensions (WxHxD)	936mm*678mm *365mm
Weight	92 Kg
Protection class	Terminal Block
CERTIFICATES	
Grid	IEC 61727
Anti-islanding	IEC 62116
Environmental	IEC 60068-2 (1-2-14-30)
Safety	IEC 62109-1/2
EMC	"EN IEC 61000-6-2 : 2019 & EN IEC 61000-6-4 : 2019"
Efficiency Measurement	IEC 61683
WARRANTY	7 Years

Note :

* ** The range of output voltage and frequency may vary depending upon different grid codes

SOLAR PV MODULE (DCR/NON-DCR) 144 HALF CUT MONO PERC CELL RANGE - NSM 525W TO 560W

High Performance Guarantee!



years
performance
warranty



years
product
warranty

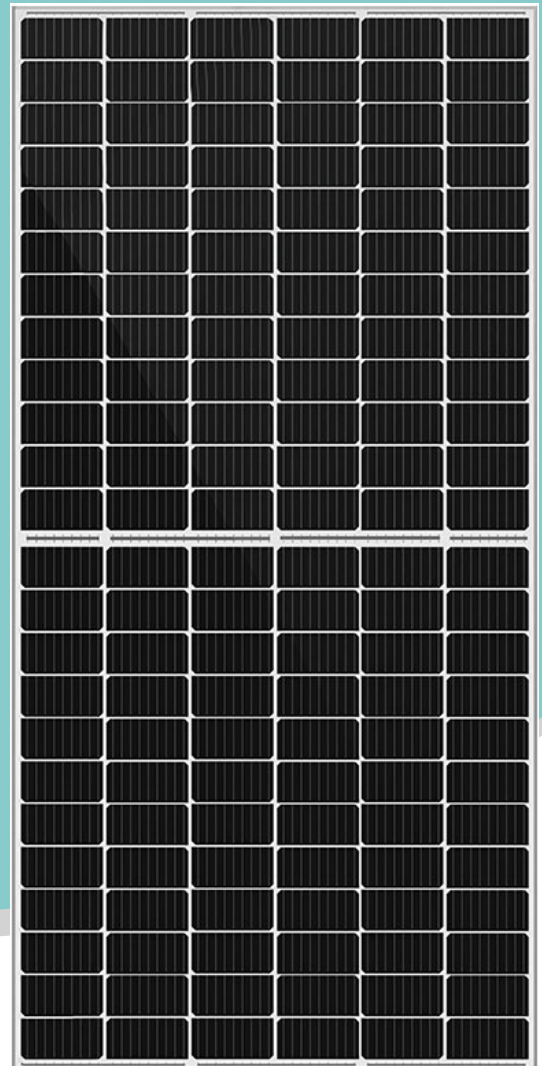
ALMM
APPROVED
LIST BY MNRE

MODULE
EFFICIENCY
UP TO 21.67%

POWER
TOLERANCE
UP TO 4.99W

Product Certification

IS 14286, IS 61730-1 & 2, IS 62804, IEC 61701, UL 61703
Fire Rating BIS ISO 9001, 14001, 45001



LEADING FEATURES



Ideal For Large Scale Installations



High Power



Better Shading Tolerance



Lower LCOE & System Cost



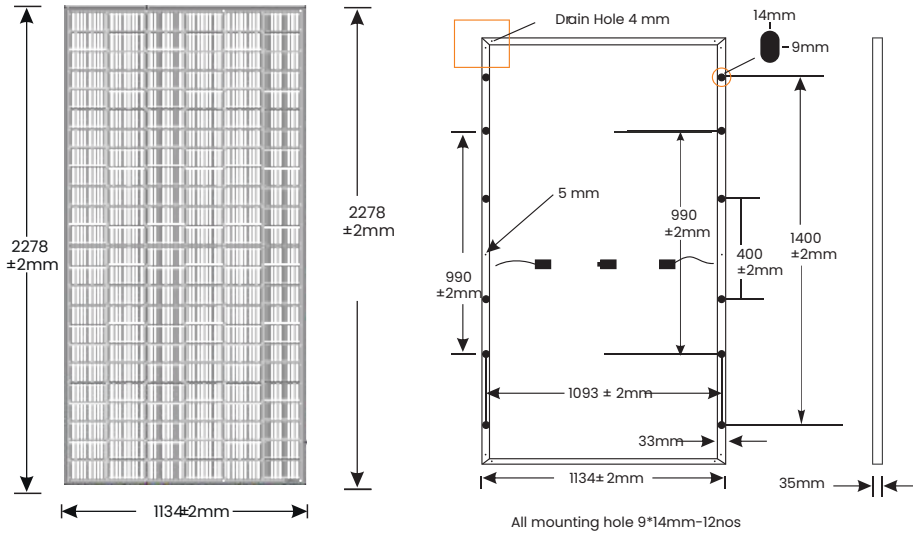
Excellent Temperature Performance



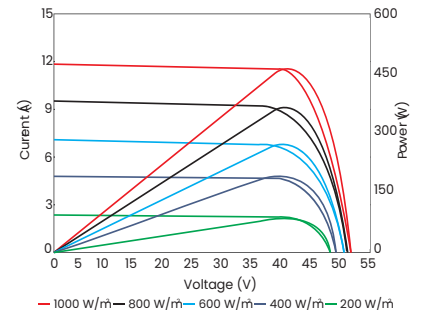
Non-destructive Cutting

Solar PV Module 144 Half Cut Mono Perc Cell

Mechanical Dimensions



I-V Curves of PV Module (for Ref)



Electrical Data

Module Type	NSM525 - 144		NSM530 - 144		NSM535 - 144		NSM540 - 144		NSM545 - 144		NSM550 - 144		NSM555 - 144		NSM560 - 144	
	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Peak Power Watts (Pmax)	525	390.46	530	393.99	535	397.67	540	401.28	545	405.17	550	408.80	555	412.53	560	416.29
Maximum Power Voltage (Vmp)	41.54	39.00	41.76	39.20	41.99	39.42	42.21	39.63	42.46	39.86	42.68	40.07	42.91	40.28	43.14	40.50
Maximum Power Current (Imp)	12.64	10.01	12.69	10.05	12.74	10.09	12.79	10.13	12.84	10.16	12.88	10.20	12.93	10.24	12.98	10.28
Open-circuit voltage (Voc)	49.80	45.77	49.94	45.92	50.0	46.06	50.16	46.20	50.3	46.35	50.49	46.51	50.62	46.66	50.75	46.81
Short-circuit current (Isc) (A)	13.45	10.74	13.5	10.77	13.56	10.81	13.60	10.84	13.65	10.88	13.69	10.91	13.74	10.94	13.78	10.98
Module Efficiency STC (%)	20.32		20.51		20.71		20.90		21.09		21.29		21.48		21.67	
Operating Temperature (°C)	-40°C ~ +85°C															
Maximum System Voltage	1500 V DC (IEC)															
Maximum series fuse rating	20A															

- ▶ **STC** Irradiance 1000 W/m² Cell Temperature 25°C AM = 1.5
- ▶ **NOCT** Irradiance 800 W/m² Ambient Temperature 20°C AM = 1.5 Wind Speed = 1 m/s

Mechanical Data

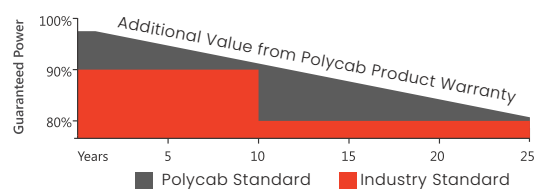
Specification	Data
Cell Type	Half cut Mono PERC
Cell Arrangement	72 Mono PERC - 144 Half Cells
Dimensions	2278x1134x35 mm
Weight	29 Kg.
Front Cover	3.2 mm ARC Glass
Frame Material	Anodized Aluminium Alloy
Junction Box	IP68 Split JB
Cable	4 mm² (IEC) - Length 0.35 mtr (Portrait)/1.4mtr (Landscape)
Connectors	MC4 Compatible
By-Pass Diodes	3 Pcs

Temperature Characteristic

Specification	Data
Temperature Co-efficient (Pmax)	-0.34% /°C
Temperature Co-efficient (Voc)	-0.26% /°C
Temperature Co-efficient (Isc)	+0.05% /°C
Nominal Operating Cell Temperature	42 ± 2°C

Performance Warranty

10 Years Product Warranty - 25 Years Linear Power Warranty



Increased Shade Tolerance

HALF-CELL MODULE

Functions like two parallel modules, enabling the half-cell string to work in partial shading



* The above data is liable to change without prior notice
Warranty applicable as per standard warranty terms

Note : The specification and key features contained in this datasheet may deviate slightly from our actual products due to the on-going innovation and product enhancement, Polycab India Ltd. reserves the right to make necessary adjustment to the information described herein at any time without further notice.

SOLAR PV MODULE (DCR/NON-DCR) 144 MONO PERC HALF CUT BIFACIAL MODULE

RANGE - NSM 525W TO 560W

High Performance Guarantee!



years
performance
warranty



years
product
warranty

MODULE
EFFICIENCY
UP TO 21.67%

POWER
TOLERANCE
UP TO 4.99W

PID
RESISTANT



LEADING FEATURES



Ideal For residential rooftop and
Large Scale Installations



High Power with Bifacial



Better Shading Tolerance



Lower LCOE & System Cost



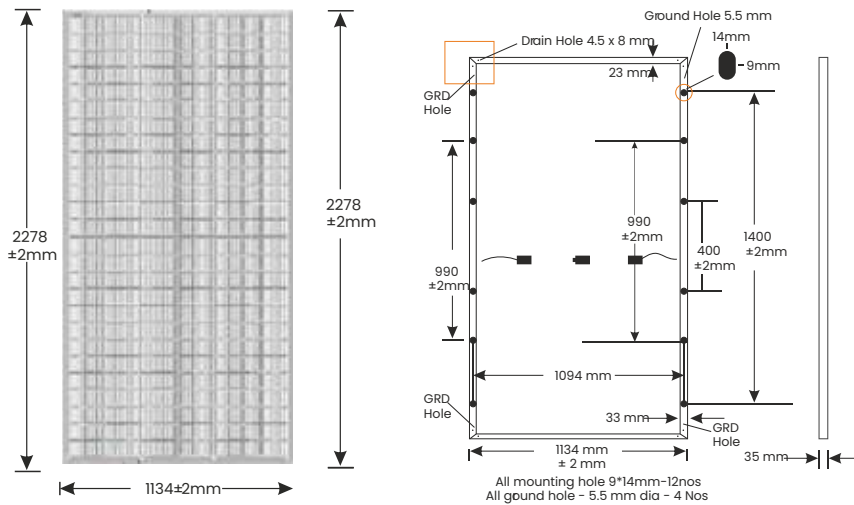
Excellent Temperature Performance



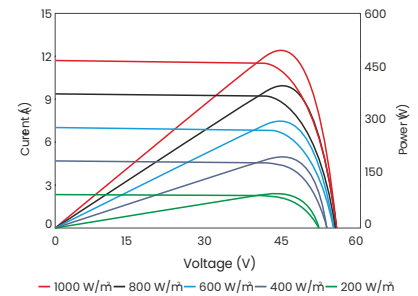
Non-destructive Cutting

Solar PV Module 144 Mono Perc Half Cut BI - FACIAL

Mechanical Dimensions



I-V Curves of PV Module (for Ref)



Degradation in the first year- 2%,
Degradation from second year onwards- 0.55%

Electrical Data

Module Type	NSM525 - 144		NSM530 - 144		NSM535 - 144		NSM540 - 144		NSM545 - 144		NSM550 - 144		NSM555 - 144		NSM560 - 144	
	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Peak Power Watts (Pmax)	525	390.46	530	393.99	535	397.67	540	401.28	545	405.17	550	408.80	555	412.53	560	416.29
Maximum Power Voltage (Vmp)	41.54	39.00	41.76	39.20	41.99	39.42	42.21	39.63	42.46	39.86	42.68	40.07	42.91	40.28	43.14	40.50
Maximum Power Current (Imp)	12.64	10.01	12.69	10.05	12.74	10.09	12.79	10.13	12.84	10.16	12.88	10.20	12.93	10.24	12.98	10.28
Open-circuit voltage (Voc)	49.80	45.77	49.94	45.92	50.0	46.06	50.16	46.20	50.3	46.35	50.49	46.51	50.62	46.66	50.75	46.81
Short-circuit current (Isc) (A)	13.45	10.74	13.5	10.77	13.56	10.81	13.60	10.84	13.65	10.88	13.69	10.91	13.74	10.94	13.78	10.98
Module Efficiency STC (%)	20.32		20.51		20.71		20.90		21.09		21.29		21.48		21.67	
Operating Temperature (°C)	-40°C ~ +85°C															
Maximum System Voltage	1500 V DC (IEC)															
Maximum series fuse rating	20A															

Bifacial Output - Backside Power Gain

Note: Measurement uncertainty ± 3%

Module Type	NSM525 - 144	NSM530 - 144	NSM535 - 144	NSM540 - 144	NSM545 - 144	NSM550 - 144	NSM555 - 144	NSM560 - 144
10 % Gain	577.5	583.0	588.5	594.0	599.5	605.0	610.5	616.0
15 % Gain	603.8	609.5	615.3	621.0	626.8	632.5	638.3	644.0
20 % Gain	630.0	636.0	642.0	648.0	654.0	660.0	666.0	672.0
25 % Gain	656.3	662.5	668.8	675.0	681.3	687.5	693.8	700.0
30 % Gain	682.5	689.0	695.5	702.0	708.5	715.0	721.5	728.0

- STC** Irradiance 1000 W/m² Cell Temperature 25°C
NOCT Irradiance 800 W/m² Ambient Temperature 20°C

- AM = 1.5
 AM = 1.5 Wind Speed = 1 m/s

Note: Bifacial gain will depend on structure height, system design and albedo.

Mechanical Data

Specification	Data
Cell Type	Half cut Mono PERC
Cell Arrangement	72 Mono PERC - 144 Half Cells
Dimensions	2278x1134x35 mm
Mounting Hole	Y - 400 / 990 / 1400 mm
Weight	29 Kg.
Front Cover	3.2 mm ARC Glass
Frame Material	Anodized Aluminium Alloy
Junction Box	IP68 Split JB
Cable	4 mm ² (IEC) - Length 0.35 mtr (Potrait)/ 1.4mtr(Landscape)
Connectors	MC4 Compatible
By-Pass Diodes	3 Pcs
Configuration	Transperant

Temperature Characteristic

Specification	Data
Temperature Co-efficient (Pmax)	-0.34% /°C
Temperature Co-efficient (Voc)	-0.26% /°C
Temperature Co-efficient (Isc)	+0.040% /°C
Nominal Operating Cell Temperature	42 ± 2°C

*The above data is liable to change without prior notice

*Warranty applicable as per standard warranty terms as available on www.navitassolar.com

Note : The specification and key features contained in this datasheet may deviate slightly from our actual products due to the on-going innovation and product enhancement, Polycab India Ltd. reserves the right to make necessary adjustment to the information described herein at any time without further notice.

POLYCAB SOLAR H1Z2Z2-K BS EN 50618

Photovoltaic Solar DC Cable, Halogen Free,
Flame Retardant

Salient Features

- Halogen free
- Electron Beam Cross-linked
- Flame retardant
- Long life
- Flexibility
- UV, Ozone resistant
- Hydrolysis resistant
- High temperature resistant



Voltage Rating

Nominal Voltage: 1500 V DC between
conductors as well as conductor and earth.
Max permitted voltage: 1800 V

Operation Temperature

Fixed: -40°C to +120°C
Maximum operating conductor
temperature: +120°C

Construction

- Conductor: Tinned copper conductor as per IEC 60228, class 5.
- Insulation: E-Beam cross linked halogen free and flame-retardant compound (XLPO)
- Sheath: E-Beam cross linked halogen free and flame-retardant compound (XLPO)

Identification

Insulation : (-ve) Black & (+ve) Red
Sheath : (-ve) Black & (+ve) Black (70%) with red
Strip (30%)

Bending Radius

For fixed installation - > 4D
For occasional movement - > 5D

Standard and References

EN/IEC 60228
EN 50618
IEC 60332-1-2

Test Voltage

6.5kV AC 50Hz

Compliance

Fire Performance
Smoke Emission

Halogen free material

Resistance to ozone
Weathering / UV

Life Expectancy
Water Resistance
-Category {(AD7/AD8)}

DIMENSIONAL CHARACTERISTICS

Single Core Cross sectional Area	Nominal insulation thickness	Nominal Sheath thickness	Approx. Overall Diameter	Max. DC Resistance at 20° C
mm ²	mm	mm	mm	Ω/km
1.5	0.7	0.8	5.0	13.7
2.5	0.7	0.8	5.5	8.21
4.0	0.7	0.8	6.0	5.09
6.0	0.7	0.8	6.5	3.39
10	0.7	0.8	7.5	1.95
16	0.7	0.9	8.5	1.24
25	0.9	1.0	10.5	0.795
35	0.9	1.1	12.0	0.565
50	1.0	1.1	14.0	0.393
70	1.1	1.2	16.0	0.277
95	1.1	1.3	18.0	0.210
120	1.2	1.3	19.5	0.164
150	1.4	1.4	21.5	0.132
185	1.6	1.6	24.5	0.108
240	1.7	1.7	27.0	0.0817
300	1.8	1.8	30.0	0.0654

CURRENT RATINGS

Nominal Cross sectional Area	Current Carrying Capacity according to method of installation		
	Single Cable Free in air	Single Cable on a surface	Two loaded cables touching, on a surface
mm ²	A	A	A
1.5	30	29	24
2.5	41	39	33
4	55	52	44
6	70	67	57
10	98	93	79
16	132	125	107
25	176	167	142
35	218	207	176
50	276	262	221
70	347	330	278
95	416	395	333
120	488	464	390
150	566	538	453
185	644	612	515
240	775	736	620
300	895	850	713

*Current Ratings are based on EN 50618 at Max. Conductor Temperature 120°C and Ambient Air temperature 60°C.

Note: the expected period of use at maximum conductor temperature at 120° C is limited to 20,000 hours

Current rating / de-rating factors other than 60°C ambient temperature.

Up to 60°C	70°C	80°C	90°C
1.00	0.92	0.84	0.75

Note: These cables can be provided with twisted formation, If required.

POLYCAB

CORPORATE OFFICE

POLYCAB INDIA LIMITED

#29, The Ruby, 21st Floor, Senapati Bapat Marg, Tulsi Pipe Road, Dadar (West),

Mumbai - 400 028, Maharashtra (India)

Ph.: +91-22-2432 7070/4, 6735 1400 | Send 'Hi' on 7304485540



solar@polycab.com



1800 267 0008



www.polycab.com

FOLLOW US ON



SCAN THE QR CODE
FOR SOLAR CATALOGUE